

MTH302 - Midsem 2

1. (5 marks) Compute

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{30}$$

explaining each step in your computation. Determine the minimal polynomial for A .

2. (3 marks) Find a matrix A whose minimal polynomial is $x^3 - 5x^2 + 6x + 8$. Justify your answer.
3. (4 marks) Let T be a triangulable operator on a finite-dimensional vector space V over a field $\mathbb{F}$. Show that for any polynomial $g(x) \in \mathbb{F}[x]$, $g(T)$ is triangulable.
4. (4 marks) Let $\mathbb{F}$ be a subfield of the complex numbers. Let V be a finite-dimensional vector space over $\mathbb{F}$. Suppose $E_1, \dots, E_k$ are projections of V and that $E_1 + \dots + E_k = I$. Prove that $E_i E_j = 0$ for $j \neq i$.
5. State True or False and give short explanations for your answers:
- (3 marks) For any $n \times n$ matrix A over a field $\mathbb{F}$, the transpose A^t has the same minimal polynomial as A .
 - (3 marks) Any two projections of the same rank are similar.
 - There exist two 4×4 matrices which have the same minimal polynomial and characteristic polynomial, but are not similar.